Wen-Nan Wu/吳文男

Postdoctoral Fellow

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Education

- 2004/09 – 2010/06: **Ph.D.** Institute of Geophysics, National Central University (NCU), Taiwan
- 1999/09 – 2001/06: **M.S.** Institute of Geophysics, NCU, Taiwan
- 1995/09 – 1999/06: **B.S.** Department of Oceanography, National Taiwan Ocean University, Taiwan

Experience

- 2024/08 – present: Postdoctoral fellow, Department of Earth Sciences, NCU, Taiwan
- 2022/03 – 2023/12: Postdoctoral fellow, Center for Environmental Studies (CES), NCU, Taiwan
- 2018/09 – 2021/12: Associate professor, College of Oceanography, Hohai University (HHU), Nanjing, China
- 2019/02 – 2020/02: Associate Director, Department of Marine Geology and Geophysics, HHU, Nanjing, China
- 2016/08 – 2018/08: Postdoctoral fellow, CES/Department of Earth Sciences, NCU, Taiwan
- 2010/09 – 2016/07: Postdoctoral fellow, Institute of Earth Sciences, Academia Sinica, Taiwan
- 2008/05 – 2009/10: Visiting student researcher, Pacific Geosciences Centre, Geological Survey of Canada, Canada

- 2004/09 – 2010/06: Graduate research assistant, Institute of Geophysics, NCU, Taiwan
- 2003/07 – 2004/08: Research assistant, Institute of Geophysics, NCU, Taiwan
- 2001/10 – 2003/06: Artillery Second Lieutenant, compulsory military service, ROC Army
- 1999/09 – 2001/06: Teaching assistant, Institute of Geophysics, NCU, Taiwan

Current Research Interests

- Delineating seismogenic structures offshore and onshore Taiwan to provide critical constraints on Taiwan orogeny and seismic hazard assessment.
- Revealing earthquake source characteristics of subduction zone earthquakes and intraplate events to better understand earthquake nucleation and triggering mechanisms, now focus on the western boundary of the Philippine Sea Plate.
- Analyzing seismicity data to comprehend earthquake processes and lithosphere deformation.

Teaching at Hohai University (2018/09 – 2021/12; outstanding teaching award, 2018-2019; 2019-2020)

- The Origin of Continents and Oceans: Introduction to Plate Tectonics (Literacy class; Fall, 2018 and 2019)
- Marine environment, resource, and climate change: Introduction to Geophysics (Introductory course for College of Oceanography freshman; Fall, 2018 and 2019)
- Introduction to geology (Undergraduate course, Fall, 2018 and 2019)
- Volcanology: Introduction to Volcano Seismology (Undergraduate course, Spring, 2019)
- Geophysics: Introduction to Seismology (Undergraduate course, Spring, 2019, 2020 and 2021)
- Marine Tectonics and Geophysics: Introduction to Seismology and Earth Structure (Graduate course; Spring, 2019, 2020 and 2021)
- Undergraduate Research (Fall, 2019)
- Undergraduate student mentor (Fall, 2019 – Fall, 2021)

Peer-Reviewed Publication

24. Wu, W.-N, J.-Y. Lin, W.-B. Doo, C.-L. Lo, and S.-K. Hsu (2024), Seismogenic structures along the deformation front from onshore to offshore SW Taiwan, *Geoscience Letters*, 11:37, doi:10.1186/s40562-024-00354-6

23. Sibuet, J-C., S. Liu, M. Zhao, **W.-N. Wu**, Y.-M. Wu, J. Cheng and J. Wu (2024), A revolution in understanding SE Asia geodynamics since 20.5 – 18 Ma, *Tectonophysics*, 884, 230397, doi:10.1016/j.tecto.2024.230397.

22. Doo, W.-B., C.-L. Lo, Y.-S. Huang, **W.-N. Wu**, and S.-Y. Wang (2023), Variations in the crustal structure and strength of plate coupling along the Ryukyu subduction zone, *Geoscience Letters*, *10*(1), 46, doi:10.1186/s40562-023-00300-y.

21. Wu, W.-N., C.-L. Lo, W.-B. Doo, J.-Y. Lin, and S.-K. Hsu (2021), Seismogenic structure along the deformation front off SW Taiwan revealed by the aftershocks of the 2017 Tainan near-shore earthquake with ocean bottom seismometers, *Tectonophysics*, 815, 228995, doi:10.1016/j.tecto.2021.228995.

20. Doo, W.-B., **W.-N. Wu**, Y.-S. Huang, C.-L. Lo, H.-F. Wang, S.-Y. Wang, and H. Kuo-Chen (2021), Deep crustal structure in the Taiwan–Ryukyu arc–trench system junction area: New constraints from gravity modelling, *Terra Nova*, 33, 407-414, doi:10.1111/ter.12525.

19. Doo, W.-B., C.-L. Lo, H. Kuo-Chen, Y.-S. Huang, **W.-N. Wu**, S.-K. Hsu and H.-F. Wang (2020), Variations in mantle lithosphere buoyancy reveal seismogenic behavior in the Sunda-Andaman subduction zone, *Geophysical Journal International*, 220, 2, 1275-1283, doi:10.1093/gji/ggz502.

18. Lo, C.-L., **W.-N. Wu**, W.-B. Doo, Y.-S. Huang, and S.-K. Hsu (2019), Crustal stress field in Ilan Plain, NE Taiwan and the westernmost Okinawa trough- deduced from seismic stress inversion, *Terrestrial Atmosphere Oceanic Sciences*, 30, 5, 613-619, doi:10.3319/TAO.2019.08.09.01.

17. Lin, J.-Y., **W.-N. Wu**, C.-C. Su, C.-H. Tsai, H.-H. Sun, Y.-F. Chen, S.-J. Chin, and Y.-C. Lin (2019), Spatial and temporal distribution of low frequency volcanic earthquakes in the southern Okinawa Trough back-arc basin, *Terrestrial Atmosphere Oceanic Sciences*, 30, 5, 621-631, doi:10.3319/TAO.2018.12.24.01.

16. Doo, W.-B., C.-L. Lo, **W.-N. Wu**, J.-Y. Lin, S.-K., Hsu, Y.-S., Huang, and H.-F. Wang, (2018), Strength of plate coupling in the southern Ryukyu subduction zone, *Tectonophysics*, 723, 223-228, doi:10.1016/j.tecto.2017.12.028.

15. Lin, J.-Y., Y.-F. Chen, C.-C. Su, S.-J., Chin, W.-B., Cheng, **W.-N. Wu**, C.-W. Liang, H.-S. Hsieh, S.-K. Hsu, and Y.-C.

Lin (2018), Seismic site response of submarine slope offshore southwestern Taiwan, *Terrestrial Atmosphere Oceanic Sciences*, 29, 1, 51-63, doi:10.3319/TAO.2017.05.09.01.

14. Wu, W.-N., Y.-T. Yen, Y.-J. Hsu, Y.-M. Wu, J.-Y. Lin and S.-K. Hsu (2017), Spatial variation of the seismogenic depth of crustal earthquakes in the Taiwan region and its implications on seismic hazard assessment, *Tectonophysics*, 708, 81-95, doi:10.1016/j.tecto.2017.04.028.

13. Wu, W.-N., C.-L. Lo and J.-Y. Lin (2017), Spatial variation of the crustal stress field in the Philippine region and its tectonic implications, *Journal of Asian Earth Sciences*, 142, 109-118, doi:10.1016/j.jseaes.2017.01.036.

12. Chin, S.-J., J.-Y. Lin, Y.-F. Chen, **W.-N. Wu** and C.-W. Liang (2016), Transition of the Taiwan-Ryukyu collision-subduction process as revealed by ocean-bottom seismometer, *Journal of Asian Earth Sciences*, 128, 149-157, doi:10.1016/j.jseaes.2016.07.008.

11. Wu, W.-N. (2015), Comment on the paper by Kavitha and Raghukanth "Regional level forecasting of seismic energy release", *Acta Geod Geophys*, 51, 773-775, doi:10.1007/s40328-015-0157-x.

10. Lin, J.-Y., **W.-N. Wu** and C.-L. Lo (2015), Megathrust earthquake potential of the Manila subduction systems revealed by the radial component of seismic moment tensor Mrr, *Terrestrial Atmosphere Oceanic Sciences*, 26, 619-630, doi:10.3319/TAO.2013.04.29.01(TC).

9. Lin, J.-Y., C.-L. Lo, **W.-N. Wu**, J.-C. Sibuet, S.-K. Hsu and Y.-Y. Wen (2015), Crustal thickening and extension induced by the Great Sumatra-Andaman earthquake of 26 December 2004: revealed by the seismic moment tensor element Mrr, *Marine Geophysical Research*, 36, 187-195, doi:10.1186/1880-5981-66-49.

8. Lin, J.-Y., J.-C. Sibuet, S.-K. Hsu, and **W.-N. Wu** (2014), Could a Sumatra-like megathrust earthquake occur in the south Ryukyu subduction zone? *Earth Planets and Space*, 66:49, doi:10.1186/1880-5981-66-49.

7. Wu, W.-N. and L. Zhao (2013), Delineating seismogenic structures by a revised best estimate method: Application to the Taiwan orogenic belt, *Journal of Seismology*, 17, 545-556, doi:10.1007/s10950-012-9336-1.

6. Wu, W.-N., L. Zhao, and Y.-M. Wu (2013), Empirical relationships between aftershock zone dimensions and moment magnitudes for plate boundary earthquakes in Taiwan, *Bulletin of the Seismological Society of America*, 103, doi:10.1785/0120120173.

5. D'Amico S., B. Orecchio, D. Presti, G. Neri, **W.-N. Wu**, I. Sandu, L. Zhu, and R. B. Hermann (2013), Source parameters of small and moderate earthquakes in the area of the 2009 L'Aquila earthquake sequences (central Italy), *Physics and Chemistry of the Earth*, 63, 77-91, doi:10.1016/j.pce.2013.02.005.

4. Lin, J.-Y. and **W.-N. Wu** (2012), Spatio-temporal distribution of seismic moment release near the source area of the 2011 Tohoku-Oki earthquake, *Earth Planets and Space*, 64, 1067-1075, doi:10.5047/eps.2012.04.006.

3. Armada, L., S.-K. Hsu, C.-Y. Ku, W.-B. Doo, **W.-N. Wu**, C. Dimalanta, and G. Yumul (2012), Possible northward extension of the Philippine Fault Zone offshore Luzon Island (Philippines), *Marine Geophysical Research*, 33, 369-377, doi:10.1007/s11001-013-9169-5.

2. Wu, W.-N., H. Kao, S.-K. Hsu, C.-L. Lo, and H.-W. Chen (2010), Spatial variation of the crustal stress field along the Ryukyu-Taiwan-Luzon convergent boundary, *Journal of Geophysical Research: Solid Earth*, 115, B11401, doi:10.1029/2009JB007080.

1. Wu, W.-N., S.-K. Hsu, C.-L. Lo, H.-W. Chen, and K.-F. Ma (2009), Plate convergence at the westernmost Philippine Sea Plate, *Tectonophysics*, 466, 162-169, doi:10.1016/j.tecto.2007.11.011.

Field Experience

- Research cruise, Ocean Bottom Seismometer (OBS) recovery South China Sea (03/2023; 22 days)
- Research cruise, OBS recovery off SW Taiwan (03/2017; 2 days)

Academic Service and Synergistic Activity

- Served as reviewer for *BSSA, EPS, GRL, GRAPY, GSAB, JAES, JGR-SE, SRL, PAG, TAO,* and *Tectonophysics*.
- Judge of Outstanding Student Paper Awards: Taiwan Geosciences Assembly **(**2011-2017)
- Serve in Editor Board: Member of the Editorial Board *Mistral Service Press* (2015 - present)
- Served as Consultant

- National Standards Review Committee number of Ocean Bottom Seismograph, Ministry of Natural Resources, China (2019)
- Workshop on Taiwan SSHAC Level 3 Project, invited by the National Center for Research on Earthquake Engineering, Taipei, 2017.
- Workshop on Seismogenesis of the Subduction Zone in South Taiwan, invited by the Institute of Nuclear Energy Research and the Institute of Geophysics, National Central University, Jhongli, 2014.
- Workshop on Seismogenesis of the Subduction Zone in North Taiwan, invited by the Institute of Nuclear Energy Research and SINOTECH Engineering Consultants Inc., Taipei, 2013.

Honor and Fellowship

- Academia Sinica Postdoctoral Research Fellowship, 2011-2012
- Outstanding Graduate Research Scholarship, National Central University, 2008
- Recipient of the Graduate Students Study Abroad Program sponsored by the National Science Council, 2008